

What a Coverage Guarantee Costs: the Price of Coverage at a Point

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Abstract

Conformal prediction promises that its intervals cover the truth nine times in ten on average over all inputs. A user usually wants that promise for the input in front of them. We show what the stronger promise costs. Suppose similar inputs have similar error distributions, and errors are spread out relative to their size. Then an interval that covers nine times in ten at a given input, on average over the data used to build it, must be wider than the interval someone who knew the true error distribution would quote. The extra width is at least of order $n^{-s/(s+d)}$ in the number n of past cases, which is $n^{-1/2}$ with one input and the simplest smoothness assumption. For exponentially distributed errors and that smoothness assumption we give a procedure that achieves it.

1 The question

What conformal prediction promises. Someone has a forecasting model and wants an interval around each forecast. Split conformal prediction sets aside n past cases, records how wrong the model was on each, and uses the 90th percentile of those errors as the half-width of every interval. The promise is that the interval contains the truth 90% of the time. The probability in that promise is an average over everything random, including which input turns up next. It is called *marginal* coverage.

What people want. A user with a particular input in hand wants to know about that input. A patient is not reassured by a rate that averages over all patients. Marginal coverage permits 99% coverage on easy inputs and 60% on hard ones, provided they average to 90%. So the natural request is coverage *at the input*.

What is known. With no assumption at all, coverage at every input is impossible with intervals of finite width (Vovk, 2012; Lei and Wasserman, 2014; Barber et al., 2021). With an assumption that the error distribution changes smoothly with the input, it becomes possible: nearby past cases are informative about this one. But it cannot be free. The interval must be wider than the ideal one that a person who knew the true error distribution would quote.

The question. How much wider? The answer depends on what, exactly, is promised at the input. The past cases are themselves a random draw, and there are two ways to treat that.

- *In expectation.* The interval at this input covers 90% of the time, counting both the luck of the next outcome and the luck of which past cases were drawn. This is what a textbook prediction interval promises.
- *With high probability.* On most possible draws of the past cases, say nine in ten, the interval built from that draw covers at least 90% of the time at this input. Nothing is averaged away: a bad draw of past cases counts as a failure.

Neither promise implies the other. The first allows an occasional draw on which coverage at this input is poor, provided the average is right. The second allows one draw in ten on which coverage is anything at all, even zero, so its average coverage can be as low as 81%. What the second promise forces is worked out in Cotton (2026a), using the theory of confidence intervals of Low (1997) and Cai and Low (2004). We have not found what the first forces in print. This note computes it.

The result. Three numbers govern the answer. The number of past cases is n . The number of input variables is d . The smoothness s says how regularly the error distribution changes with the input: $s = 1$ means it changes at a bounded rate, and larger s means smoother. All three are defined precisely in Section 2.

Table 1 lists each promise with what it forces. An entry such as $n^{-1/2}$ means a quantity proportional to $1/\sqrt{n}$, so a hundred times more data makes it ten times smaller. The second row is the contribution of this note.

The rows are not measured in the same way, and the difference matters. For the promise in expectation we bound the *average* threshold over draws. For the promise on most draws the known result bounds the threshold on a *typical* draw: on more than half of draws it exceeds the ideal by the stated amount. The average would be the wrong measure there, because that promise lets a procedure quote zero on a few draws, which lowers its average threshold without breaking the promise.

Promise at the input	Forced above the ideal	$s = d = 1$	Source
None (averaged over inputs)	nothing		Lei and Wasserman (2014)
Averaged over draws	average threshold, by $n^{-s/(s+d)}$	$n^{-1/2}$	this note
On most draws	typical threshold, by $n^{-s/(2s+d)}$	$n^{-1/3}$	Cotton (2026a)
On every draw	threshold, by a fixed amount*		Cotton (2026a)
With no smoothness assumed	infinite width		Barber et al. (2021)

Table 1: What each promise forces. The first row says the marginal promise forces nothing: a procedure can keep it and still approach the ideal interval as fast as the ideal interval can be estimated. The second and third promises are not ordered, neither implies the other, and their costs are measured differently. The fourth implies both. *When the error size is known to lie in a bounded range. With no such bound the fourth promise forces an infinite threshold.

An example with numbers. The example is fully specified as follows.

- *Inputs.* One input variable, taking values between 0 and 1. The n past cases have evenly spaced inputs $x_i = (i - \frac{1}{2})/n$, and the input we care about is $x_0 = \frac{1}{2}$. Even spacing is used so that the numbers are exact. The theorems treat inputs placed at random.

- *Scores.* The *score* of a past case is the size of the model’s error on it. Given the input x , the score is exponentially distributed with mean $e^{\theta(x)}$, where θ is a function of the input that the procedure does not know. This is a choice made so that every number below is exact and not simulated. It is the distribution of the absolute error when errors follow a Laplace distribution. Writing the mean as e^θ keeps it positive and puts θ on a log scale, so that a change of 0.1 in θ means errors about 10% larger.
- *The truth.* $\theta(x) = 0$ for every x . So every score, at every input, is exponential with mean one. The procedure is not told this.
- *What the procedure is told.* That θ changes by at most 1 per unit distance. In the notation of Section 2 this is smoothness $s = 1$ with constant $L = 1$. The procedure must keep its promise for every such θ , not only the true one.
- *Target.* 90% coverage at x_0 , so $\alpha = 0.1$.

The *oracle threshold* is the 90th percentile of an exponential with mean one, which is $\log 10 = 2.303$. Someone who knew the truth would quote exactly that, and nothing smaller covers 90% of the time.

A *threshold* is the number a procedure quotes as the largest error it expects at x_0 , so the interval is the forecast plus or minus the threshold. A procedure that does not know the truth quotes a threshold that depends on the data, so we report its average over draws of the scores. Both procedures below multiply the average score of the k nearest past cases by a constant, as in Section 7, with k chosen to make the average threshold as small as possible.

Promise at x_0	$n = 10,000$		$n = 1,000,000$	
	average threshold	above oracle	average threshold	above oracle
Oracle, knows the truth	2.303		2.303	
In expectation	2.338	1.5%	2.306	0.15%
On 90% of draws	2.502	8.7%	2.344	1.8%

The in-expectation procedure uses $k = 148$ neighbours at $n = 10,000$ and the other uses $k = 564$. With a hundred times more data the first excess falls by a factor of ten, which is $100^{1/2}$. The second falls by a factor of 4.8, close to $100^{1/3} = 4.6$. The theorems below show that no procedure keeping the first promise can improve on the “in expectation” row by more than a constant factor.

The last row needs a caution. It is the average threshold of one procedure that keeps the second promise by padding on every draw. A procedure may instead quote zero on 8% of draws and pad the rest a little more. It still keeps the second promise, and at $n = 1,000,000$ its average threshold is 2.170, below the oracle. So the last row is what a sensible procedure costs, not what the promise forces on the average. No coin is needed for this. A procedure can quote zero whenever the first past case has an input below 0.08, which happens on 8% of draws when inputs are placed at random and tells it nothing about x_0 .

Why the two prices differ. Any threshold computed from data wobbles from one draw of the data to the next. Suppose it wobbles evenly above and below the ideal value. Under the in-expectation promise the draws that land too high nearly cancel the draws that land too low.

They do not cancel exactly, because landing too low loses more coverage than landing equally too high gains. What is left over is proportional to the *square* of the wobble.

The reason is that coverage is a curved function of the threshold. Raising the threshold from t to $t + w$ adds the probability that the score lands between them, and scores are scarcer the further out one goes, so each further step adds less than the one before. In the example above the oracle threshold 2.303 covers 90%. A threshold 0.5 higher covers 93.9%, a gain of 3.9 points. A threshold 0.5 lower covers 83.5%, a loss of 6.5 points. A procedure that quotes each half the time is right on average about the threshold and covers 88.7%, short by 1.3 points. Halve the wobble to 0.25 and the shortfall is 0.31 points, a quarter of what it was. A straight line would give no shortfall at all, and the first departure of a smooth curve from its tangent line is quadratic. To restore 90% the procedure must raise its thresholds by an amount proportional to the shortfall, hence to the square of the wobble. Figure 1 shows the three thresholds.

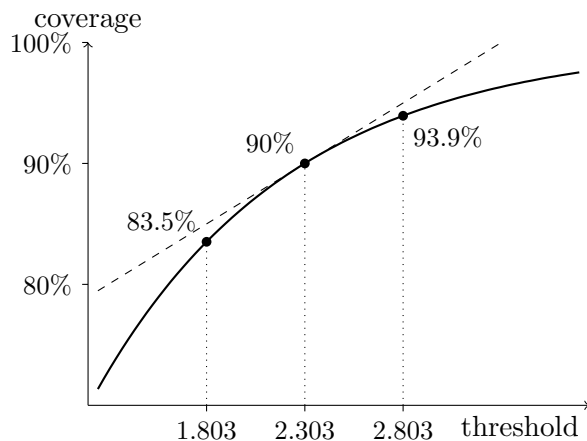


Figure 1: Coverage as a function of the threshold, for scores that are exponential with mean one. The solid curve bends downward. The dashed line is the straight line touching it at the oracle threshold 2.303. Moving 0.5 below the oracle loses 6.5 points of coverage, and moving 0.5 above gains 3.9.

Under the high-probability promise nothing cancels. On each of the draws that count, the threshold must itself be high enough, so it is padded by the wobble and not by its square. A squared small number is much smaller, so the in-expectation promise can tolerate more wobble. It therefore borrows from fewer, closer past cases, which reduces the error from treating neighbours as identical.

Why this matters for conformal prediction. It separates two statements that are often run together. Conformal prediction’s own guarantee is nearly free and says nothing about any particular input. Every guarantee that does say something about a particular input has a price, the price can be written down, and it is paid in width.

How the note is organized. Section 2 defines every term. Section 3 states the two results in words. Section 4 states the assumptions and says when each fails. Section 5 proves that no procedure can do better than the stated rate. Section 6 checks the assumptions in the model of the example and gives numerical constants. Section 7 gives a procedure that achieves the rate. Section 8 says what the result means, what it does not show, and what is borrowed from earlier

work. A reader who wants the idea and not the proofs can stop after Section 3 and resume at Section 8.

2 Setup and definitions

Inputs, outcomes, scores. An *input* is a vector $x \in \mathbb{R}^d$. A forecasting model has already been fitted on other data and is not changed again, and we call it *frozen*. For an input X with outcome Y , the *score* $R \geq 0$ measures how wrong the frozen model was, for example $R = |Y - \hat{\mu}(X)|$ where $\hat{\mu}$ is the model's forecast. Large scores are large errors.

Calibration sample. The *calibration sample* $D_n = \{(X_i, R_i)\}_{i=1}^n$ is a set of n past cases, not used to fit the model. The pairs are independent draws from one distribution. The inputs X_i have probability density g on $\mathbb{R}^d$. The positions of the inputs $X_1, \dots, X_n$ are called the *design*.

How the score distribution depends on the input. We describe the distribution of the score at input x by one number $\theta(x)$, such as the log of the typical error size there. Formally there is a family of distribution functions $\{F_u\}$ indexed by a real number u , and

$$\mathbb{P}(R \leq t \mid X = x) = F_{\theta(x)}(t).$$

The function $\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ is unknown. We call it the *field*. A single parameter is enough for a lower bound: if the true family has more parameters, a lower bound proved for a one-parameter slice of it still holds.

Test point. The *test point* x_0 is a fixed input chosen in advance. It is not random.

Procedure, threshold, prediction set. A *procedure* is a rule that turns the calibration sample into a number $T \geq 0$, the *threshold*. The *prediction set* for the score at x_0 is $[0, T]$: the procedure asserts that the model's error at x_0 will be at most T . For the absolute error score this is the interval $\hat{\mu}(x_0) \pm T$. The rule may also flip coins. We write $T = T(D_n, U)$ with U the coin flips, independent of the data.

Miscoverage level and oracle threshold. Fix $\alpha \in (0, 1)$, the permitted failure rate, so that $1 - \alpha$ is the target coverage. With $\alpha = 0.1$ the target is 90%. The *oracle threshold* is the threshold someone who knew the true distribution at x_0 would quote,

$$q_\alpha(u) = F_u^{-1}(1 - \alpha) \quad \text{at } u = \theta(x_0),$$

the $(1 - \alpha)$ quantile of the score at x_0 . No threshold below it achieves the target.

The guarantee. Write $\mathbb{E}_\theta$ and $\mathbb{P}_\theta$ for expectation and probability over the calibration sample and the coin flips when the field is θ . For a realized threshold T , the chance that a fresh score at x_0 falls below it is $F_{\theta(x_0)}(T)$. This is the coverage achieved by that particular T , and it is random because T is.

Definition 1 (Validity at a point in expectation). Let Θ be a set of fields. A procedure is *valid* at x_0 in expectation over Θ if

$$\mathbb{E}_\theta[F_{\theta(x_0)}(T)] \geq 1 - \alpha \quad \text{for every } \theta \in \Theta. \tag{1}$$

Equivalently $\mathbb{P}_\theta(R_0 \leq T) \geq 1 - \alpha$ where R_0 is a fresh score at x_0 , independent of the calibration sample and the coin flips, and the probability runs over both R_0 and the calibration sample. The requirement must hold for every field in Θ , because the procedure does not know which one is true.

For comparison, *marginal* validity replaces the fixed x_0 by a random input X_0 drawn from g and averages over it as well. *High-probability* validity at level $1 - \eta$ asks that $\mathbb{P}_\theta\{F_{\theta(x_0)}(T) \geq 1 - \alpha\} \geq 1 - \eta$. Validity at x_0 for every x_0 implies marginal validity. High-probability validity with $\eta > 0$ does not imply validity in expectation: it guarantees only $\mathbb{E}_\theta F_{\theta(x_0)}(T) \geq (1 - \eta)(1 - \alpha)$. Nor does validity in expectation imply high-probability validity.

The cost. The *excess* of a procedure under the field θ is how far its threshold sits above the oracle's on average,

$$\Delta_\theta(T) = \mathbb{E}_\theta[T] - q_\alpha(\theta(x_0)),$$

with $\Delta_\theta = \infty$ if $\mathbb{E}_\theta T = \infty$. For an interval $\hat{\mu}(x_0) \pm T$ it is half the expected extra width. We ask how small the excess can be made for a valid procedure. Since the procedure does not know the field, we look at its excess in the worst case over the fields it must allow for, written $\sup_{\theta \in \Theta} \Delta_\theta(T)$.

Smoothness. The assumption that makes the problem possible is that the field does not change abruptly. For $0 < s \leq 1$ and $L > 0$, the field θ is *Hölder smooth of order s with constant L* if

$$|\theta(x) - \theta(y)| \leq L |x - y|^s \quad \text{for all } x, y.$$

With $s = 1$ this says θ changes by at most L per unit distance. For $s > 1$, write k for the largest integer strictly below s . The condition is that θ has derivatives up to order k and that every derivative of order k satisfies the display above with exponent $s - k$. We write $\mathcal{H}^s(L)$ for the set of such fields and call it the *Hölder ball*. Larger s means smoother, and larger L means rougher.

Bandwidth. Any procedure that uses past cases near x_0 must decide how near. The radius h of the neighbourhood it uses is the *bandwidth*. A neighbourhood of radius h holds about $n g(x_0) h^d$ past cases, up to a constant. A wide neighbourhood holds many cases, so averages over it are steady, but those cases are less like x_0 . The error from treating neighbours as if they sat at x_0 is the *bias*. For $s \leq 1$ it is at most Lh^s . For $s > 1$ the field may have a slope, so a plain average of neighbours can be off by more, and a bias of order Lh^s is reached only after fitting and removing the slope and the higher terms up to order k .

A distance between distributions. For two probability distributions P and Q , with P having density dP/dQ relative to Q , the *chi-square divergence* is

$$\chi^2(P \parallel Q) = \int \left(\frac{dP}{dQ} \right)^2 dQ - 1.$$

It is zero when $P = Q$ and grows as they separate. It measures how hard it is to tell from data which of the two produced it. It is not symmetric in P and Q .

3 Results

The first result says validity is expensive in a specific way.

Theorem 2 (Lower bound, informal). *Suppose past cases are neither absent nor piled up near x_0 (Assumption 1), and the scores are spread out relative to their size, as exponential or half-normal scores are (Assumption 2). Let θ_0 be a constant field, so that the score distribution is the same at every input. Then for all large n , every procedure valid at x_0 in expectation over $\mathcal{H}^s(L)$ has*

$$\Delta_{\theta_0}(T) \geq c L^{d/(s+d)} (n g_+)^{-s/(s+d)},$$

where g_+ is an upper bound on the density of inputs near x_0 , so that $n g_+$ measures how many past cases fall near x_0 , and $c > 0$ does not depend on n or L .

The excess is forced when the truth is the simplest field in the class, the one with no variation at all. The procedure pays for bumps that are not there, because it cannot rule them out.

The precise version is Corollary 7. The second result says the bound cannot be improved.

Theorem 3 (Attainment, informal). *When scores are exponentially distributed with a mean that varies smoothly with the input, the log of that mean is known to lie in a bounded range, and $s \leq 1$, there is an explicit procedure that is valid at x_0 in expectation for every n , with no approximation, and whose excess is at most $C L^{d/(s+d)} (n g_-)^{-s/(s+d)}$ under every field in the class. Here g_- is a lower bound on the density of inputs near x_0 and C does not depend on n or L .*

The precise version is Proposition 10. The two bounds have the same dependence on n and on L and differ only in their constants. Together they give the second row of Table 1.

The factor $L^{d/(s+d)}$ says that a rougher class costs more. If $L = 0$ the field is known to be constant, every past case is as good as a case at x_0 , and the bound vanishes.

4 Assumptions

Assumption 1 (The input density is bounded near the test point). There are a radius $r_0 > 0$ and constants $0 < g_- \leq g_+ < \infty$ with $g_- \leq g(x) \leq g_+$ for all x within distance r_0 of x_0 .

This says past cases near x_0 are neither absent nor piled up. It fails at the edge of the data and in sparse regions.

Assumption 2 (The score family is regular near a baseline). There are a baseline value u_0 , a width $\varepsilon_0 > 0$ and positive constants $K, c_q, C_q, \bar{f}, \kappa, \rho$ such that for all $u \in [u_0, u_0 + \varepsilon_0]$:

- (a) *Nearby parameters are hard to tell apart.* $\chi^2(F_u \| F_{u_0}) \leq K(u - u_0)^2$.
- (b) *The oracle threshold moves with the parameter.* $c_q(u - u_0) \leq q_\alpha(u) - q_\alpha(u_0) \leq C_q(u - u_0)$.
- (c) *The distribution function bends downward at the oracle threshold.* F_u has a density, its value at the oracle threshold $f_u := F'_u(q_\alpha(u))$ satisfies $0 < f_u \leq \bar{f}$, and for every $t \geq 0$

$$F_u(t) \leq 1 - \alpha + f_u(t - q_\alpha(u)) - \kappa \psi_\rho(t - q_\alpha(u)), \quad \psi_\rho(v) := \min\{v^2, \rho|v|\}. \quad (2)$$

The constants have these roles. K says how fast two score distributions become distinguishable as their parameters separate. The divergence is proportional to the *square* of the separation because that is how every smooth family behaves for small separations. c_q and C_q say how fast the oracle threshold moves when the parameter moves. $\bar{f}$ bounds the score density at the oracle threshold. κ and ρ describe the curvature of the distribution function and are explained next.

Part (c) carries the argument. The first two terms on the right of (2) are the straight line touching the graph of F_u at the oracle threshold. The inequality says the graph lies below that line, by an amount that is quadratic in the distance v from the threshold when v is small and linear when v is large. The function ψ_ρ is that amount, and ρ is where it changes from quadratic to linear. It is the mathematical form of “landing too low loses more than landing too high gains”.

The condition is required for every $t \geq 0$, including $t = 0$, and that makes it a restriction on the family. At $t = 0$ the left side is zero, and with $q = q_\alpha(u)$ the condition reads $q f_u \leq 1 - \alpha - \kappa \psi_\rho(q)$. In particular it needs $q f_u < 1 - \alpha$. This holds for scores that are spread out relative to their size and fails for scores concentrated near a positive value. With $\alpha = 0.1$ the product $q f_u$ is 0.23 for the exponential, 0.34 for the absolute value of a normal, and 0.86 for a gamma distribution with shape 20, all of which qualify. It is 1.03 for a gamma with shape 30 and 1.15 for a Weibull with shape 5, which do not. A decreasing density at the threshold is not enough.

The restriction is not an artefact of the proof. For the gamma with shape 30 the oracle threshold is 37.199, and a procedure that quotes 0 with small probability and 37.80 otherwise covers 90% of the time with an average threshold of 37.155. That is below the oracle. So for concentrated scores the excess as we have defined it can be negative, and a different measure of cost is needed.

Assumption 3 (The class contains a small local bump). The set of fields is $\Theta = \mathcal{H}^s(L)$. There is a function $b : \mathbb{R}^d \rightarrow [0, 1]$, the *bump profile*, with $b(0) = 1$ and $b(z) = 0$ for $|z| \geq 1$, and there are constants $a \in (0, 1]$ and $h_0 \in (0, r_0]$, such that for every $0 < h \leq h_0$ the two fields

$$\theta_0(x) = u_0 \quad \text{and} \quad \theta_1(x) = u_0 + a L h^s b\left(\frac{x - x_0}{h}\right)$$

both belong to Θ . Write $\|b\|_2^2 = \int b(z)^2 dz$.

The field θ_0 is flat. The field θ_1 equals θ_0 except within distance h of x_0 , where it rises to a peak of height $a L h^s$ at x_0 . A bump of width h can be at most about $L h^s$ tall and still be Hölder smooth, which is why the height has that form. The constant a is whatever is needed to keep the bump inside the ball. For $s = d = 1$ the triangle $b(z) = \max\{1 - |z|, 0\}$ works with $a = 1$. Figure 2 shows the two fields.

5 The lower bound

Plan. The two fields θ_0 and θ_1 differ only near x_0 , so a calibration sample drawn under one looks much like a sample drawn under the other. A valid procedure must cover under both. Under θ_1 the oracle threshold is higher, so the procedure’s threshold must be higher on average. It cannot meet that requirement with rare enormous thresholds, because coverage cannot exceed one: beyond a certain size a larger threshold buys nothing. And a procedure cannot behave very differently on samples it cannot tell apart, unless its threshold varies a great deal. Lemma 4 shows validity itself limits how much the threshold may vary. Together these force the excess up under the flat field.

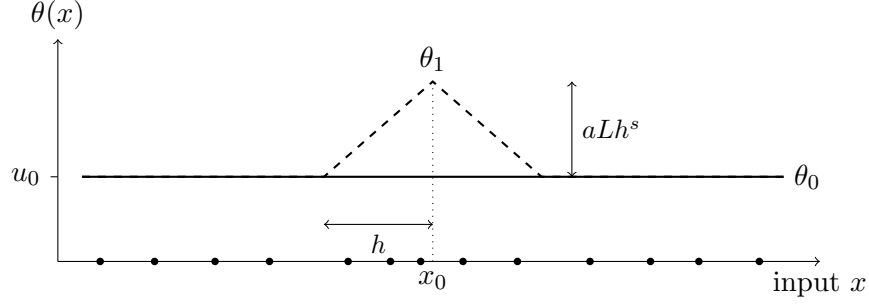


Figure 2: The flat field θ_0 (solid) and the bump field θ_1 (dashed). Dots are the inputs of past cases. Only the few cases under the bump carry any information about which field is true. A narrower bump is lower and covers fewer cases, so it is harder to detect.

Lemma 4 (Validity bounds the spread by the excess). *Let Assumption 2(c) hold at $u = \theta(x_0)$, write $q = q_\alpha(u)$, and let T satisfy (1) under θ with $\mathbb{E}_\theta T < \infty$. Then*

$$\Delta_\theta \geq 0 \quad \text{and} \quad \mathbb{E}_\theta \psi_\rho(T - q) \leq \frac{\bar{f}}{\kappa} \Delta_\theta.$$

In words: a valid threshold sits above the oracle on average, and its typical squared distance from the oracle is at most a constant times its excess.

Proof. Put $t = T$ in (2) and take expectations. The left side is at least $1 - \alpha$ by (1), and $\mathbb{E}_\theta(T - q) = \Delta_\theta$, so $1 - \alpha \leq 1 - \alpha + f_u \Delta_\theta - \kappa \mathbb{E}_\theta \psi_\rho(T - q)$. That is $\kappa \mathbb{E}_\theta \psi_\rho(T - q) \leq f_u \Delta_\theta$. The left side is nonnegative and $f_u > 0$, so $\Delta_\theta \geq 0$. Then $f_u \Delta_\theta \leq \bar{f} \Delta_\theta$ gives the second claim. $\square$

Lemma 5 (Change of expectation). *Let P_0 and P_1 be probability distributions with P_1 having a density relative to P_0 , and let Y be a random variable with finite variance under P_0 . Writing $\mathbb{E}_j$ and Var_j for expectation and variance under P_j ,*

$$(\mathbb{E}_1 Y - \mathbb{E}_0 Y)^2 \leq \chi^2(P_1 \| P_0) \text{Var}_0(Y).$$

In words: if two distributions are hard to tell apart, the average of any quantity can differ between them only if that quantity is highly variable.

Proof. With $\Lambda = dP_1/dP_0$ we have $\mathbb{E}_0 \Lambda = 1$, so $\mathbb{E}_1 Y - \mathbb{E}_0 Y = \mathbb{E}_0[(\Lambda - 1)(Y - \mathbb{E}_0 Y)]$. The Cauchy-Schwarz inequality bounds the square of this by $\mathbb{E}_0(\Lambda - 1)^2 \cdot \text{Var}_0(Y)$, and $\mathbb{E}_0(\Lambda - 1)^2 = \chi^2(P_1 \| P_0)$. This is Lemma 2.1 of Derumigny and Schmidt-Hieber (2023). $\square$

Theorem 6 (Two-point bound at the flat field). *Let Assumptions 1–3 hold. Fix a bump width $h \in (0, h_0]$, let $\delta = aLh^s$ be the bump height, and suppose*

$$\delta \leq \varepsilon_0, \quad C_q \delta \leq \rho/2, \quad I := n K g_+ \|b\|_2^2 \delta^2 h^d \leq 1. \quad (3)$$

Then every procedure satisfying (1) under both θ_0 and θ_1 has

$$\Delta_{\theta_0}(T) \geq \min \left\{ A_1 L h^s, \frac{A_2}{n g_+ h^d} \right\}, \quad (4)$$

with, writing $q_0 = q_\alpha(u_0)$,

$$A_1 = \frac{c_q a}{2}, \quad A_2 = \frac{\kappa c_q^2}{8 \bar{f} C_{\text{cap}} K \|b\|_2^2}, \quad C_{\text{cap}} = \max \left\{ 1, \frac{q_0}{\rho}, \frac{1}{2} + \frac{\alpha}{\kappa \rho^2} \right\}.$$

The constants A_1 , A_2 and C_{cap} depend on the score family, the target α and the bump profile. They do not depend on n , L or h .

The three conditions in (3) say the bump is small enough for Assumption 2 to apply, small enough that the two oracle thresholds are close, and small enough that the two fields are hard to tell apart from n observations. The quantity I measures the last: it is the bump height squared times the number of past cases under the bump. The two terms on the right of (4) are the bias a bandwidth h can hide, and the reciprocal of the number of past cases within h of x_0 .

Proof. Write P_j for the distribution of the sample and coin flips (D_n, U) under θ_j , and $\mathbb{E}_j$ for expectation under it. Write $q_1 = q_\alpha(u_0 + \delta)$ for the oracle threshold under the bump field, $\gamma = q_1 - q_0$, $F_1 = F_{u_0 + \delta}$ and $f_1 = f_{u_0 + \delta}$. Since $b(0) = 1$ we have $\theta_1(x_0) = u_0 + \delta$, and Assumption 2(b) gives $c_q \delta \leq \gamma \leq C_q \delta \leq \rho/2$. Write $\Delta_0 = \Delta_{\theta_0}$. If $\Delta_0 = \infty$ there is nothing to prove, so assume $\mathbb{E}_0 T < \infty$. Let $v = T - q_0$.

Step 1: cap the threshold where coverage saturates. Dropping the last term of (2) gives $F_1(T) \leq 1 - \alpha + f_1(T - q_1) = 1 - \alpha + f_1(v - \gamma)$. Also $F_1(T) \leq 1$. Put

$$B = \gamma + \frac{\alpha}{f_1}, \quad Z = \min\{v, B\}.$$

Then, for every value of T ,

$$F_1(T) \leq \min \{1, 1 - \alpha + f_1(v - \gamma)\} = 1 - \alpha + f_1(Z - \gamma).$$

Taking expectations under the bump field and using (1), $1 - \alpha \leq 1 - \alpha + f_1(\mathbb{E}_1 Z - \gamma)$, so

$$\mathbb{E}_1 Z \geq \gamma. \tag{5}$$

The variable Z is bounded, since $-q_0 \leq Z \leq B$, so no assumption on $\mathbb{E}_1 T$ is needed.

Step 2: the density at the threshold exceeds $\kappa\rho$. For $t > q_1 + \rho$ the right side of (2) is $1 - \alpha + (f_1 - \kappa\rho)(t - q_1)$. The left side tends to $1 > 1 - \alpha$ as $t \rightarrow \infty$. So $f_1 > \kappa\rho$, and with $\gamma \leq \rho/2$,

$$\frac{B}{\rho} \leq \frac{1}{2} + \frac{\alpha}{\kappa\rho^2} \leq C_{\text{cap}}.$$

Step 3: under the flat field, Z has small mean and small variance. Since $Z \leq v$,

$$\mathbb{E}_0 Z \leq \mathbb{E}_0 v = \Delta_0. \tag{6}$$

We claim $Z^2 \leq C_{\text{cap}} \psi_\rho(v)$. Note $B > 0$. If $|v| \leq \rho$, then $|Z| \leq |v|$ and $Z^2 \leq v^2 = \psi_\rho(v)$. If $v < -\rho$, then $Z = v$ and $|v| \leq q_0$ because $T \geq 0$, so $Z^2 = |v|^2 \leq q_0 |v| = (q_0/\rho) \psi_\rho(v)$. If $v > \rho$, then $0 < Z \leq B$ and $Z \leq v$, so $Z^2 \leq Bv = (B/\rho) \psi_\rho(v)$. In each case the constant is at most C_{cap} . Lemma 4 under θ_0 now gives

$$\text{Var}_0(Z) \leq \mathbb{E}_0 Z^2 \leq \frac{\bar{f} C_{\text{cap}}}{\kappa} \Delta_0. \tag{7}$$

Step 4: combine. From (5) and (6), $\mathbb{E}_1 Z - \mathbb{E}_0 Z \geq \gamma - \Delta_0$. Either $\Delta_0 \geq \gamma/2$, in which case $\Delta_0 \geq c_q \delta/2 = A_1 L h^s$ and (4) holds. Or $\Delta_0 < \gamma/2$, in which case $\mathbb{E}_1 Z - \mathbb{E}_0 Z > \gamma/2 > 0$, and Lemma 5 with (7) gives

$$\frac{\gamma^2}{4} \leq \chi^2(P_1 \| P_0) \frac{\bar{f} C_{\text{cap}}}{\kappa} \Delta_0.$$

Step 5: the two fields are hard to tell apart. The coin flips U have the same distribution under both fields and are independent of the sample, so they do not affect the divergence. For a single pair (X, R) the input has the same density g under both fields, so the divergence between the two distributions of the pair is the average over inputs of the divergence between the two score distributions:

$$\chi_{(1)}^2 := \int g(x) \chi^2(F_{\theta_1(x)} \| F_{u_0}) dx \leq K \delta^2 \int b\left(\frac{x - x_0}{h}\right)^2 g(x) dx \leq K \delta^2 g_+ h^d \|b\|_2^2.$$

The first inequality is Assumption 2(a), which applies because $\theta_1(x) - u_0 = \delta b((x - x_0)/h)$ lies between 0 and $\delta \leq \varepsilon_0$. The second is Assumption 1, which applies because the bump vanishes beyond distance $h \leq r_0$ from x_0 , followed by the substitution $z = (x - x_0)/h$. For n independent pairs the quantity $1 + \chi^2$ multiplies: $1 + \chi^2(P_1 \| P_0) = (1 + \chi_{(1)}^2)^n \leq e^{n \chi_{(1)}^2} \leq e^I$. Since $I \leq 1$ and $e^x - 1 \leq 2x$ on $[0, 1]$, $\chi^2(P_1 \| P_0) \leq 2I$.

Step 6: conclude. With $\gamma \geq c_q \delta$,

$$\Delta_0 \geq \frac{\kappa \gamma^2}{4 \bar{f} C_{\text{cap}} \cdot 2I} \geq \frac{\kappa c_q^2 \delta^2}{8 \bar{f} C_{\text{cap}} n K g_+ \|b\|_2^2 \delta^2 h^d} = \frac{A_2}{n g_+ h^d}.$$

The bump height δ cancels. □

Corollary 7 (The rate). *Under Assumptions 1–3 there is n_0 such that for $n \geq n_0$ every procedure valid at x_0 in expectation over $\mathcal{H}^s(L)$ has, at the flat field $\theta_0 \equiv u_0$,*

$$\Delta_{\theta_0}(T) \geq \min\{A_1, A_2\} L^{d/(s+d)} (n g_+)^{-s/(s+d)}.$$

Proof. Theorem 6 holds for every admissible width h , so we may choose one. A convenient choice is $h_* = (L n g_+)^{-1/(s+d)}$, where $L h^s$ and $1/(n g_+ h^d)$ are equal, with common value $L^{d/(s+d)} (n g_+)^{-s/(s+d)}$. The right side of (4) is then at least $\min\{A_1, A_2\}$ times that value. The width that makes the right side largest equates $A_1 L h^s$ with $A_2/(n g_+ h^d)$ and gives the constant $A_1^{d/(s+d)} A_2^{s/(s+d)}$ in place of $\min\{A_1, A_2\}$. It remains to check (3). The height $\delta = a L h_*^s$ tends to zero as n grows. And $I = K \|b\|_2^2 a^2 (n g_+ h_*^d \cdot L h_*^s) (L h_*^s) = K \|b\|_2^2 a^2 L h_*^s$, because the first bracket equals one by the choice of h_* . So I tends to zero too. Hence (3) and $h_* \leq h_0$ hold for all n beyond some n_0 . Both θ_0 and θ_1 belong to $\mathcal{H}^s(L)$, so a procedure valid over the class is valid under both. □

Remark 8 (Rare enormous thresholds do not help). A procedure might try to satisfy the bump field by quoting a huge threshold on a small fraction of samples. Step 1 is what defeats this. Once a threshold is large enough that the tangent line has reached one, making it larger adds nothing to coverage, and Z stops increasing. So the requirement under the bump field is a requirement on the capped quantity Z , whose variance under the flat field is controlled by the flat field's own excess.

6 A model where everything is explicit

The assumptions of Section 4 are abstract. This section checks them in the model of the example, so that the lower bound becomes a number.

Let the score at input x be exponentially distributed with mean $e^{\theta(x)}$. So $F_u(t) = 1 - e^{-t/m}$ with $m = e^u$. Write $\ell = \log(1/\alpha)$. The oracle threshold is $q_\alpha(u) = \ell e^u$ and the density there is $f_u = \alpha e^{-u}$. With $\alpha = 0.1$ and $u = 0$ the oracle threshold is $\ell = 2.303$, the figure in the example of Section 1.

Proposition 9. *Take $u_0 = 0$ and $\varepsilon_0 = \frac{1}{2}$. Assumption 2 holds with*

$$K = 3, \quad c_q = \ell, \quad C_q = 2\ell(e^{1/2} - 1), \quad \bar{f} = \alpha, \quad \kappa = \alpha e^{-2}, \quad \rho = e^{1/2}.$$

Proof. (a) Let $r = e^u$ with $u \in [0, \frac{1}{2}]$, so $1 \leq r < 2$. The density of F_u is $r^{-1}e^{-t/r}$ and that of F_0 is e^{-t} . Then $\int_0^\infty (r^{-1}e^{-t/r})^2 e^t dt = r^{-2}/(2/r - 1) = 1/(r(2 - r))$, finite because $r < 2$. So $\chi^2(F_u \| F_0) = 1/(r(2 - r)) - 1 = (r - 1)^2/(r(2 - r))$. The ratio of this to u^2 increases on $(0, \frac{1}{2}]$ and equals 2.906 at $u = \frac{1}{2}$.

(b) $q_\alpha(u) - q_\alpha(0) = \ell(e^u - 1)$. The function $e^u - 1$ is convex and vanishes at zero, so on $[0, \frac{1}{2}]$ it lies above its tangent u and below the chord $2(e^{1/2} - 1)u$.

(c) Write $v = t - q$ and $z = v/m$, where $q = \ell m$. Then $e^{-t/m} = \alpha e^{-z}$ and the amount by which the tangent line exceeds the distribution function is

$$1 - \alpha + f_u(t - q) - F_u(t) = \alpha(z - 1 + e^{-z}) =: \alpha \varphi(z).$$

We claim $\varphi(z) \geq e^{-1} \min\{z^2, |z|\}$ for every real z . For $z \leq 0$: by Taylor's theorem $e^{-z} = 1 - z + z^2/2 - z^3 e^{-\xi}/6$ for some ξ between 0 and z , and the last term is nonnegative, so $\varphi(z) \geq z^2/2$. For $0 < z \leq 1$: the ratio $\varphi(z)/z^2$ is decreasing and equals e^{-1} at $z = 1$. For $z \geq 1$: the ratio $\varphi(z)/z = 1 - (1 - e^{-z})/z$ is increasing and equals e^{-1} at $z = 1$. Next, $\min\{z^2, |z|\} = \min\{v^2/m^2, |v|/m\}$ decreases as m increases, so for every $m \leq e^{1/2}$ it is at least $\min\{v^2/e, |v|/e^{1/2}\} = e^{-1} \min\{v^2, e^{1/2}|v|\}$. Here e is Euler's number and v is the distance from the threshold. Therefore the gap is at least $\alpha e^{-2} \min\{v^2, e^{1/2}|v|\}$, which is (2) with $\kappa = \alpha e^{-2}$ and $\rho = e^{1/2}$. Finally $f_u = \alpha e^{-u} \leq \alpha$. $\square$

With $\alpha = 0.1$, $s = d = 1$, $L = 1$, the triangular bump and a uniform input density, the constants are $C_{\text{cap}} = \frac{1}{2} + e = 3.22$, $A_1 = 1.15$ and $A_2 = 0.0139$.

How far apart the two bounds are. Theorem 6 with the best choice of h gives an excess of at least $\sqrt{A_1 A_2}/\sqrt{n} = 0.13/\sqrt{n}$. At $n = 10,000$ that is 0.0013. The procedure in the example has excess 0.035 there, which is $3.5/\sqrt{n}$. So the way the excess shrinks with n is settled, and the constant in front is known only to within a factor of about 28. The lower bound's constant is the loose one: the proof gives away factors at each step.

7 A procedure that attains the rate

Stay in the exponential model and take $0 < s \leq 1$. Assume the fields take values in a bounded interval $[\underline{u}, \bar{u}]$ and write $W = \bar{u} - \underline{u}$ for its length. Nothing is assumed about the inputs far from x_0 . The lower bound of Corollary 7 applies to this smaller class as well, provided $\underline{u} \leq u_0 < \bar{u}$, so that the bump, which rises above u_0 , fits inside the range once h is small. The bound uses only the two fields θ_0 and θ_1 .

The procedure. Choose a number k of neighbours. Let N_k be the k past cases whose inputs are closest to x_0 , let ρ_k be the distance from x_0 to the farthest of them, and let $\bar{R}_k$ be the average of their scores. Quote

$$T_k = c_k e^{\beta_k} \bar{R}_k, \quad \beta_k = \min\{L\rho_k^s, W\}, \quad c_k = k(\alpha^{-1/k} - 1). \quad (8)$$

The average $\bar{R}_k$ estimates the mean score near x_0 . The factor e^{β_k} pads for the worst case in which the neighbours have smaller errors than x_0 does. Two field values cannot differ by more than $L\rho_k^s$, by smoothness, nor by more than W , by the bounded range, and β_k is the smaller of the two. The factor c_k converts a mean into a $(1 - \alpha)$ threshold and pads for the noise in the average. It exceeds ℓ by about $\ell^2/(2k)$, which is the variance term.

Proposition 10 (Exact validity and the rate). *For every $\theta \in \mathcal{H}^s(L)$ with values in $[\underline{u}, \bar{u}]$ and every $n \geq k \geq 1$, whatever the positions of the inputs, a fresh score R_0 at x_0 satisfies $\mathbb{P}_\theta(R_0 \leq T_k | X_1, \dots, X_n) \geq 1 - \alpha$. Under Assumption 1, with k of order $L^{-d/(s+d)}(ng_-)^{s/(s+d)}$ and n large,*

$$\sup_{\theta} \Delta_\theta(T_k) \leq C L^{d/(s+d)} (ng_-)^{-s/(s+d)},$$

the supremum being over the same fields, with C depending on α, d, W and $\bar{u}$ but not on n or L .

Proof. Validity. Fix the positions of the inputs. The scores in N_k are then independent exponentials with means $m_i = e^{\theta(X_i)}$, and R_0 is an independent exponential with mean $m_0 = e^{\theta(x_0)}$. An exponential with mean m_0 exceeds a value v with probability e^{-v/m_0} , and an exponential R_i with mean m_i has $\mathbb{E}e^{-cR_i} = 1/(1 + cm_i)$. So for a constant $\lambda > 0$,

$$\mathbb{P}(R_0 > \lambda \bar{R}_k) = \mathbb{E} \exp(-\lambda \bar{R}_k/m_0) = \prod_{i \in N_k} \left(1 + \frac{\lambda m_i}{k m_0}\right)^{-1}.$$

Each neighbour is within ρ_k of x_0 , so Hölder smoothness gives $\theta(X_i) \geq \theta(x_0) - L\rho_k^s$, and the bounded range gives $\theta(X_i) \geq \theta(x_0) - W$. Hence $m_i/m_0 \geq e^{-\beta_k}$. With $\lambda = c_k e^{\beta_k}$ each factor is at most $(1 + c_k/k)^{-1}$, and the product is at most $(1 + c_k/k)^{-k} = \alpha$ by the choice of c_k .

Excess. Given the positions, $\mathbb{E}\bar{R}_k = k^{-1} \sum m_i \leq m_0 e^{\beta_k}$, so $\mathbb{E}T_k \leq c_k m_0 e^{2\beta_k}$. Since $e^x - 1 \leq x + x^2 e^x/2$ for $x \geq 0$, $c_k = k(e^{\ell/k} - 1) \leq \ell + \ell^2 e^\ell/(2k)$. With $q = \ell m_0$,

$$\mathbb{E}[T_k | X_1, \dots, X_n] - q \leq m_0 \left\{ \ell(e^{2\beta_k} - 1) + \frac{\ell^2 e^\ell}{2k} e^{2\beta_k} \right\}.$$

Since $0 \leq \beta_k \leq W$ and the exponential is convex, $e^{2\beta_k} - 1 \leq C' \beta_k$ with $C' = (e^{2W} - 1)/W$, and $e^{2\beta_k} \leq e^{2W}$. The fields are bounded, so $m_0 \leq e^{\bar{u}}$. It remains to average β_k over the positions of the inputs. We use

$$\beta_k \leq L \min\{\rho_k, r_0\}^s + W \mathbf{1}\{\rho_k > r_0\}.$$

Let v_d be the volume of the unit ball in $\mathbb{R}^d$. For $t \leq r_0^d$ the ball of radius $t^{1/d}$ about x_0 receives each input with probability at least $p = g_- v_d t$, by Assumption 1. The event $\rho_k^d > t$ says fewer than k of the n inputs fell in that ball. Put $t_0 = 2k/(ng_- v_d)$ and suppose $t_0 \leq r_0^d$. For $t_0 \leq t \leq r_0^d$ the expected count np is at least $2k$, and the Chernoff bound for a count falling below half its expectation gives $\mathbb{P}(\rho_k^d > t) \leq \exp(-ng_- v_d t/8)$. Therefore

$$\mathbb{E} \min\{\rho_k, r_0\}^d = \int_0^{r_0^d} \mathbb{P}(\rho_k^d > t) dt \leq t_0 + \frac{8}{ng_- v_d} \leq \frac{10k}{ng_- v_d}, \quad \mathbb{P}(\rho_k > r_0) \leq e^{-ng_- v_d r_0^d/8}.$$

Since $s \leq d$, Jensen's inequality gives $\mathbb{E} \min\{\rho_k, r_0\}^s \leq (\mathbb{E} \min\{\rho_k, r_0\}^d)^{s/d}$. Therefore

$$\sup_{\theta} \Delta_{\theta}(T_k) \leq C \left\{ L \left(\frac{k}{ng_-} \right)^{s/d} + \frac{1}{k} + W e^{-ng_- v_d r_0^d/8} \right\}.$$

The stated k makes the first two terms equal, it satisfies $t_0 \leq r_0^d$ for large n , and the third term is exponentially small. $\square$

Validity here holds for every arrangement of the inputs and every n . The bound on the excess averages over arrangements.

8 Reading the result

Variance, not standard deviation. With bandwidth h there are about ngh^d past cases in play. The in-expectation promise forces the average threshold up by $Lh^s + 1/(ngh^d)$: the bias a neighbourhood of that size can hide, plus the *variance* of an average over that many cases. A procedure that keeps the high-probability promise by padding on every draw pads by $Lh^s + (ngh^d)^{-1/2}$, with the *standard deviation* in place of the variance, which gives $n^{-s/(2s+d)}$. The exponent $s/(s+d)$ lies strictly between $s/(2s+d)$ and $2s/(2s+d)$.

The two promises are not rungs of one ladder. The high-probability promise does not imply the in-expectation promise, and it does not force a positive excess on average. A procedure may spend part of its permitted failure rate on quoting zero. The known lower bound for that promise is a statement about the threshold on a majority of draws (Low, 1997; Cotton, 2026a). So the comparison between the second and third rows of Table 1 is between two different promises, each with the measure of cost that suits it. It is not a statement that one promise is a cheaper version of the other.

Undersmoothing. The best bandwidth for validity in expectation is $(Lng)^{-1/(s+d)}$. The best bandwidth for estimating the score distribution accurately is $(L^2ng)^{-1/(2s+d)}$, which is wider. So coverage at a point in expectation is bought by using fewer, closer neighbours than an estimator would.

What is not shown. The lower bound holds for every smoothness $s > 0$ and every score family satisfying the assumptions. The matching procedure is given only for the exponential model with $s \leq 1$. For $s > 1$, and for other families, whether the rate can be attained is open. The constant C in Proposition 10 depends on the range W but not on L . The results are for independent past cases with randomly placed inputs.

The fan. The coverage $F_{\theta(x_0)}(T)$ achieved by a realized threshold is random, and its distribution over calibration samples is the *fan* studied in Cotton (2026b). The two promises are statements about that distribution. Validity in expectation constrains its mean. High-probability validity constrains its lower tail. A fan built from m cases has standard deviation of order $m^{-1/2}$ and variance of order $1/m$, and here m is the number of past cases within the bandwidth. That is where the two prices come from. That paper shows that dependence among the calibration scores changes the width of the fan. We have not worked out what that does to the prices here, which assume independent past cases.

The classical parallel. When there are no inputs at all, the same distinction separates the two classical kinds of tolerance limit. The standard limit that contains a fraction β of the population on average over samples sits above the true quantile by an amount of order $1/n$. The standard limit that contains that fraction on most samples sits above it by an amount of order $n^{-1/2}$. The present result is the version with inputs and smoothness.

What is borrowed. Lemma 5, and the trade-off between bias and variance that it drives, are due to Low (1995) and Derumigny and Schmidt-Hieber (2023). That coverage balances bias against variance is observed for confidence intervals by Calonico et al. (2022).

What is new. Lemma 4, which converts a coverage requirement into a limit on the variance of the threshold. Theorem 6 and Corollary 7, which price coverage at a point for prediction sets, with randomly placed inputs in any dimension and the dependence on L . The procedure (8), which is exactly valid. And Table 1, which puts the promises side by side with their prices.

References

- Cotton, P. (2026a). Locally honest coverage forces nonparametric inflation: a sharpness lower bound for prediction sets. Working note.
- Cotton, P. (2026b). The width of the conformal fan. Working paper.
- Barber, R. F., Candès, E. J., Ramdas, A. and Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference* 10(2), 455–482.
- Cai, T. T. and Low, M. G. (2004). An adaptation theory for nonparametric confidence intervals. *Annals of Statistics* 32(5), 1805–1840.
- Calonico, S., Cattaneo, M. D. and Farrell, M. H. (2022). Coverage error optimal confidence intervals for local polynomial regression. *Bernoulli* 28(4), 2998–3022.
- Derumigny, A. and Schmidt-Hieber, J. (2023). On lower bounds for the bias-variance trade-off. *Annals of Statistics* 51(4), 1510–1533.
- Lei, J. and Wasserman, L. (2014). Distribution-free prediction bands for non-parametric regression. *Journal of the Royal Statistical Society, Series B* 76(1), 71–96.
- Low, M. G. (1995). Bias-variance tradeoffs in functional estimation problems. *Annals of Statistics* 23(3), 824–835.
- Low, M. G. (1997). On nonparametric confidence intervals. *Annals of Statistics* 25(6), 2547–2554.
- Vovk, V. (2012). Conditional validity of inductive conformal predictors. *Proceedings of the Asian Conference on Machine Learning*, PMLR 25, 475–490.